

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275671

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Brechbiel; Scott E. et al.

MEDICAL DEVICE ACTUATORS AND RELATED METHODS

Abstract

A medical device may comprise a shaft extending from a proximal end to a distal end. A control wire may extend through a lumen of the shaft and to an end effector at the distal end of the shaft. The medical device may further comprise an operation portion coupled to the shaft. The operation portion may further include a housing, one or more attachment devices for coupling the medical device to another medical device, an actuation mechanism for actuating the end effector. The actuation mechanism may include a lever extending through an opening of a pivotable body within the housing. The control wire may be coupled to the pivotable body.

Inventors: Brechbiel; Scott E. (Arlington, MA), Schaubhut; Andrew James (Bolton, MA)

Applicant: Boston Scientific Scimed, Inc. (Maple Grove, MN)

Family ID: 95365214

Assignee: Boston Scientific Scimed, Inc. (Maple Grove, MN)

Appl. No.: 19/066877

Filed: February 28, 2025

Related U.S. Application Data

us-provisional-application US 63648353 20240516

us-provisional-application US 63560007 20240301

Publication Classification

Int. Cl.: A61B1/00 (20060101); A61B1/015 (20060101)

U.S. Cl.:

CPC A61B1/00133 (20130101); A61B1/015 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/648,353, filed on May 16, 2024 and U.S. Provisional Patent Application Ser. No. 63/560,007, filed on Mar. 1, 2024, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] Various aspects of this disclosure relate generally to medical device actuators and related methods. In particular, aspects of this disclosure relate to actuation mechanisms for accessory devices coupled to other medical devices, among other aspects.

BACKGROUND

[0003] During endoscopic procedures, accessory devices may be used along with scope devices. Each of the accessory device and the scope device may have one or more actuators to be activated by an operator. Conventionally, different operators have operated the accessory device and the scope device. For example, a physician may actuate, operate and/or maintain a position of a scope device (e.g., a duodenoscope) while a technician may actuate and/or operate an accessory device (e.g., a tome). The use of two operators may lead to, e.g., delayed responses, miscues, and/or increased risks during the procedure.

SUMMARY

[0004] Each of the aspects disclosed herein may include one or more aspects of the features described in connection with any of the other disclosed aspects.

[0005] In an aspect, a medical device may comprise a shaft extending from a proximal end to a distal end. A control wire may extend through a lumen of the shaft and to an end effector at the distal end of the shaft. The medical device may further comprise an operation portion coupled to the shaft. The operation portion may further include a housing, one or more attachment devices for coupling the medical device to another medical device, an actuation mechanism for actuating the end effector. The actuation mechanism may include a lever extending through an opening of a pivotable body within the housing. The control wire may be coupled to the pivotable body.

[0006] Any of the devices, systems, or methods disclosed herein may include any of the following features, alone or in any combination. The medical device may further include a port assembly configured to couple the shaft to the operation portion. The port assembly may be rotatable relative to the operation portion. The port assembly may include two channels in fluid communication with the lumen of the shaft. One channel of the two channels may be configured to receive the control wire. Another channel of the two channels may be configured to couple to a fluid source. The pivotable body may include a protrusion configured to align with the one channel of the two channels. The housing may include a first slot on an outer surface of the housing and a second slot inside the housing. The lever may extend through the first slot and the second slot. The lever may be movable along the first slot and the second slot, and relative to the housing. The first slot and the second slot may extend parallel to one another. The pivotable body may include a channel configured to receive the control wire. The channel may include a first portion and a second portion extending approximately parallel to one another. The channel may include a third portion extending between and approximately perpendicular to the first portion and the second portion. The pivotable

body may include a protrusion. The second portion of the channel may extend along the protrusion. The control wire may be fixed to the channel. The channel may include a first end and a second end. A second end of the channel may be configured to align with the lumen of the shaft. The actuation mechanism may be a first actuation mechanism configured to move the control wire in a proximal direction or a distal direction. The medical device may further include a second actuation mechanism configured to rotate the control wire. The second actuation mechanism may include a first wheel rotatably coupled to a second wheel. The control wire may extend through the first wheel and the second wheel may be coupled to a knob.

[0007] In another example, a medical device may comprise a shaft. A control wire may extend through a lumen of the shaft and to an end effector at a distal end of the shaft. The medical device may further comprise an operation portion coupled to the shaft. The operation portion may include a housing configured to be coupled to a handle of a scope device and an actuation mechanism coupled to the control wire for actuating the end effector. The actuation mechanism may include an actuator, a first arm, and a second arm movably coupled to one another.

[0008] Any of the devices, systems, and methods disclosed herein may include any of the following features, alone or in any combination. A first end of the first arm may be coupled to the actuator and a second end of the first arm may be coupled to a first end of the second arm. A second end of the second arm may be coupled to the control wire. The first arm may be configured to move in an axial direction and the second arm may be configured to rotate.

[0009] In a further aspect, a medical device may comprise a shaft including an end effector at a distal end of the shaft. A control wire may extend through the shaft and may be coupled to the end effector. The medical device may further comprise an operation portion including a housing configured to couple to a handle of another medical device and an actuation mechanism for actuating the end effector. The actuation mechanism may include a knob, a rod, and a pair of wheels. The rod may extend from the knob to a first wheel of the pair of wheels. The control wire may extend through and may be rotatably fixed to a second wheel of the pair of wheels. A rotational axis of the first wheel may be transverse to a rotational axis of the second wheel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are incorporated in and constitute a part of this application, illustrate exemplary aspects of this disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0011] FIGS. 1A and 1B illustrate an exemplary medical device, in accordance with some aspects of this disclosure.

[0012] FIGS. 2A-2D illustrate features of an exemplary medical system including the medical device of FIGS. 1A and 1B coupled to an exemplary accessory device in a first configuration (FIGS. 2A and 2B) and in a second configuration (FIGS. 2C and 2D).

[0013] FIGS. 3A-3C illustrate features of an exemplary insertion portion of the accessory device of FIGS. 2A-2D in various configurations.

[0014] FIG. 4 illustrates an exemplary alternative actuator of the accessory device of FIGS. 2A-2D.

[0015] FIGS. 5A and 5B illustrate features of an exemplary medical system including the medical device of FIGS. 1A and 1B coupled to an exemplary accessory device.

[0016] FIGS. 6A and 6B illustrate features of an exemplary medical system including the medical device of FIGS. 1A and 1B coupled to an exemplary accessory device.

[0017] FIGS. 7A and 7B illustrate features of an exemplary medical system including the medical device of FIGS. 1A and 1B coupled to an exemplary accessory device.

[0018] FIGS. 8A-8C illustrate features of an exemplary medical system including the medical

device of FIGS. 1A and 1B coupled to an exemplary accessory device.

[0019] FIG. 9 illustrates an exemplary alternative actuation mechanism for an accessory device.

[0020] FIG. 10 illustrates features of an exemplary medical system including the medical device of FIGS. 1A and 1B coupled to an exemplary accessory device.

[0021] FIG. 11 illustrates features of an exemplary medical system including the medical device of FIGS. 1A and 1B coupled to an exemplary accessory device.

[0022] FIG. 12 illustrates features of an exemplary accessory device.

[0023] FIGS. 13-15 illustrate various exemplary actuators of the accessory devices of FIGS. 11 and 12.

DETAILED DESCRIPTION

[0024] Particular aspects of the present disclosure are described in greater detail below. The terms and definitions provided herein control, if in conflict with terms and/or definitions incorporated by reference.

[0025] The terms “proximal” and “distal” are used herein to refer to the relative positions of the components of exemplary medical devices. As used herein, “proximal” refers to a position relatively closer to the exterior of the body or closer to an operator using the medical device. In contrast, “distal” refers to a position relatively further away from the operator using the medical device, or closer to the interior of the body. In various of the Figures, arrows labeled “P” and “D” illustrate proximal and distal directions.

[0026] As used herein, the terms “comprises,” “comprising,” “including,” “includes,” “having,” “has,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. The term “exemplary” is used in the sense of “example,” rather than “ideal.” Relative terms such as “about,” “substantially,” and “approximately,” etc., are used to indicate a possible variation of $\pm 10\%$ of the stated numeric value or range. Ordinals such as “first,” “second,” and the like are used for convenience of referring to elements and do not imply an ordered relationship.

[0027] Although duodenoscopes are referenced herein for illustration purposes, it will be appreciated that the disclosure encompasses any suitable medical device configured to allow an operator to access and view internal body anatomy of a subject (e.g., patient) and/or to deliver medical instruments or accessory devices, such as, for example, biopsy forceps, graspers, baskets, snares, probes, scissors, retrieval devices, lasers, and/or other tools, into the subject's body. The medical devices herein may be inserted into a variety of body lumens and/or cavities, such as, for example, the urinary tract or gastrointestinal tract. It will be appreciated that, unless otherwise specified, endoscopes, bronchoscopes, gastroscopes, endoscopic ultrasonography (“EUS”) scopes, colonoscopes, ureteroscopes, laparoscopes, cystoscopes, aspiration scopes, sheaths, catheters, or any other suitable delivery device or medical device may be used in connection with the features described herein.

[0028] Features of the medical systems herein may enable a single operator to control and/or actuate two or more medical devices either simultaneously or independently, eliminating the need for a second operator to assist with operating the devices during a medical procedure. According to some aspects of the present disclosure, the medical system may include a medical device (e.g., a duodenoscope) coupled to an accessory device. The accessory device may include one or more attachment devices to facilitate coupling the accessory device to the medical device. The accessory device may include an operation portion and an insertion portion that is sized and shaped to be received within a working lumen of the medical device. The insertion portion may be coupled to the operation portion via a rotatable port assembly including two or more channels in fluid communication with the operation portion and the insertion portion. The operation portion may include one or more actuation mechanisms for actuating a distal end of the insertion portion and/or

actuating an end effector (e.g., a tome, basket, needle, snare, clip, brush, forceps, etc.) at the distal end of the insertion portion. For example, the one or more actuation mechanisms may be configured to, e.g., rotate the end effector, articulate the end effector, and/or control other aspects of the end effector.

[0029] Features of the medical systems herein may facilitate fine tune control of the medical device, the accessory device, and other medical devices (e.g., a syringe) by a single operator during a medical procedure. For example, in addition to a single operator controlling the medical device and the accessory device during a medical procedure, such as Endoscopic Retrograde Cholangiopancreatography (ERCP), it may be beneficial to have the single operator control a syringe for delivering fluid to a target site (e.g., an ampulla) to better manage pressure at the target site. Features of the medical systems herein may improve the ergonomics and/or usability of the medical device and the accessory device. For example, the operation portion of the accessory device may include, e.g., recesses, protrusions, textured surfaces, smooth surfaces, and/or markers to facilitate ease of use, among other aspects.

[0030] Reference will now be made in detail to examples of the present disclosure described above and illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

[0031] FIG. 1A depicts an exemplary medical device **110** (e.g., a duodenoscope) having a handle **112** and an insertion portion **114**. FIG. 1B shows a proximal end of handle **112**. Medical device **110** may also include an umbilicus **116** for purposes of connecting medical device **110** to sources of, e.g., air, water, suction, power, etc., as well as to image processing and/or viewing equipment. Insertion portion **114** may include a sheath or shaft **118** and a distal tip **120**. Distal tip **120** may include an imaging device **122** (e.g., a camera) and a lighting source **124** (e.g., an LED or an optical fiber). In some examples, as shown in FIG. 1A, distal tip **120** may be side-facing (i.e., side viewing). That is, imaging device **122** and lighting source **124** may face radially outward, perpendicularly, approximately perpendicularly, or otherwise transverse to a longitudinal axis of shaft **118** and distal tip **120**.

[0032] Distal tip **120** may also include an elevator **126** for changing an orientation of a medical instrument inserted in a working lumen of medical device **110**. Elevator **126** may alternatively be referred to as a swing stand, pivot stand, raising base, or any suitable other term. Elevator **126** may be rotatable via, e.g., an actuation wire or another control element that extends from handle **112**, through shaft **118**, to elevator **126**.

[0033] A distal portion of shaft **118** that is connected to distal tip **120** may have a steerable section **128**. Steerable section **128** may include, e.g., an articulation joint. Shaft **118** and steerable section **128** may include a variety of structures which are known or may become known in the art.

[0034] Handle **112** may have one or more actuators/control mechanisms **130**. One or more of control mechanisms **130** may provide control over steerable section **128**. One or more of control mechanisms may allow for provision of air, water, suction, etc. For example, handle **112** may include control knobs **132**, **134** for left, right, up, and/or down control of steerable section **128**. For example, one of knobs **132**, **134** may provide left/right control of steerable section **128**, and the other of knobs **132**, **134** may provide up/down control of steerable section **128**. Handle **112** may further include a first locking mechanism **136** and a second locking mechanism **142** (e.g., knobs or levers) for preventing steering and/or braking of steerable section **128** in at least one of an up, down, left, or right direction. Handle **112** may include a control lever **138** (see FIG. 1B). Control lever **138** may raise and/or lower elevator **126**, via connection between control lever **138** and an actuating wire that extends from control lever **138**, through shaft **118**, to elevator **126**. A port **140** may allow passage of a medical instrument through port **140**, into a working lumen of the medical device **110**, through shaft **118**, to distal tip **120**.

[0035] Although FIG. 1A depicts a side-viewing medical device **110** (a device having imaging device **122** and/or lighting source **124** facing substantially radially outward), it will be appreciated

that medical device **110** may alternatively be a forward-viewing device (a device having imaging device **122** and/or lighting source **124** facing substantially distally). In aspects, medical device **110** may include both a side-viewing imaging device **122** and/or lighting source **124** and a forward-facing imaging device **122** and/or lighting source **124**.

[0036] FIGS. 2A-2D illustrate internal and external features of a medical system **200** that includes medical device **110** (e.g., a first medical device) removably coupled to an accessory device **201** (e.g., a second medical device). Broken lines indicate internal features. Accessory device **201** may include an operation portion **241** having one or more attachment devices for removably coupling accessory device **201** to medical device **110**. For example, operation portion **241** may include a strap **205a** and a clip **205b** for securing operation portion **241** to a distal portion **148** of handle **112**. Strap **205a** may be configured to wrap at least partially around a body **144** of handle **112**. Clip **205b** may be configured to clip onto sides of port **140**, which may prevent rotation and/or movement of operation portion **241** relative to handle **112**. Clip **205b** may be integral with operation portion **241**. Alternatively, clip **205b** may be separate piece(s) from operation portion **241**. Although strap **205a** and clip **205b** are described, it will be appreciated that operation portion **241** may utilize other suitable attachment devices (e.g., other straps or clips, or clamps, pincers, bands, etc.) for securing accessory device **201** to medical device **110**.

[0037] Accessory device **201** may further include a port assembly **207** (e.g., a Y-port assembly) coupled to operation portion **241**. Port assembly **207** may include a first port **209** defining a first channel **219a** and a second port **211** defining a second channel **219b**. First port **209** may include a proximal portion **237** coupled to operation portion **241** and a distal portion **246** coupled to an insertion portion **250** of accessory device **201** (shown in FIGS. 3A-3C). For example, proximal portion **237** of first port **209** may be received within an opening formed on a housing **225** (e.g., at a distal end **295** of housing **225**) of operation portion **241** to secure port assembly **207** to operation portion **241**, while allowing port assembly **207** to rotate about a central longitudinal axis A of first port **209**. First channel **219a** may be aligned with and/or otherwise in fluid communication with one or more lumens or cavities of housing **225**, e.g., a lumen **290**.

[0038] Distal portion **246** of first port **209** may be coupled to a shaft **252** of insertion portion **250** (shown in FIGS. 3A-3C). Shaft **252** may extend distally from port assembly **207** and to a distal end **254** having an end effector **256**. Shaft **252** may be sized and shaped to be received within port **140** and the working lumen of shaft **118** of medical device **110**. In aspects, shaft **252** may define a lumen in fluid communication with first channel **219a** and/or second channel **219b**. Second port **211** may be configured to couple to a syringe **213** or other fluid source for delivering fluid through second channel **219b**, through the lumen of shaft **252**, and to a target site. Alternatively, second port **211** may allow passage of a medical instrument (e.g., a guidewire) through second channel **219b**, through the lumen of shaft **252**, and to a target site.

[0039] Accessory device **201** may include a control wire **281** (see, e.g., FIG. 2D) extending distally from within housing **225** (e.g., through lumen **290**), through first channel **219a** of first port **209**, through the lumen of shaft **252**, and to end effector **256** for actuating end effector **256**, which will be described in greater detail below. In FIGS. 3A-3C, end effector **256** is a tome. As shown in FIGS. 3A-3C, wire **281** may include a distal end fixed to a distal end of end effector **256**. Although a tome is referenced herein for illustration purposes, it will be appreciated that distal end **254** may include any suitable end effector, e.g., a basket, needle, snare, clip, brush, forceps, stent, grasper, stapler, suturing device, etc. In examples, shaft **252** may further include a braid, coil, or other structure to enable rotation of shaft **252** and/or end effector **256**.

[0040] Although port assembly **207** is described as having two ports, it will be appreciated that port assembly **207** may include any number of ports, e.g., one port or three ports. For example, port assembly **207** may include first port **209** for receiving wire **281**, second port **211** for receiving fluid, and a third port for receiving a guidewire. In addition, although shaft **252** is described as having one lumen, it will be appreciated that shaft **252** may include any number of lumens, e.g., two or

more lumens. For example, shaft **252** may include a first lumen in communication with first channel **219a** for receiving wire **281** and a second, separate lumen in communication with second channel **219b** for receiving fluid.

[0041] Operation portion **241** may include an actuation mechanism **260** for actuating end effector **256**. Actuation mechanism **260** may include a lever **203** (e.g., an actuator) that may be operable to pivot or rotate a pivotable member **217**. Lever **203** may have a substantially straight shape. Wire **281** may extend proximally from end effector **256**, through shaft **252**, through first channel **219a** of first port **209**, through housing **225** of operation portion **241**, and to actuation mechanism **260**.

[0042] For example, wire **281** may be coupled to pivotable member **217**. A portion of wire **281** may be fixed to pivotable member **217** such that movement of pivotable member **217** may cause movement of wire **281**. Pivotable member **217** may include a body **233** positioned within a cavity **227** of housing **225**. Body **233** may have a substantially triangular shape (or a shape like a sector of a circle), and may include a first edge **245a**, a second edge **245b**, and a substantially curved, third edge **245c**. First edge **245a** may have a complementary shape to a shape of a first wall **249a** of cavity **227** and curved edge **245c** may have a complementary shape to a shape of a curved, second wall **249b** of cavity **227**. Body **233** may be pivotable about a pivot point **231** in a direction G or in a direction H (shown in FIG. 2C but applicable in all of FIGS. 2A-2D). Pivot point **231** may include an axle, such as a pin, rivet, or similar structure that may be received within a ring **263** of body **233**, such that pivotable member **217** is rotationally/pivotably movable about pivot point **231**. Body **233** may further include a protrusion **289** extending substantially perpendicular to second edge **245b** and away from first edge **245a**. An outer surface of protrusion **289** may be continuous with third edge **245c**. For example, a portion of third edge **245c** may define a portion of protrusion **289**.

[0043] Body **233** may include a channel **221** adjacent second edge **245b** for receiving a portion of wire **281**. As shown in FIG. 2D, channel **221** may include a first portion **221a**, a second portion **221b**, and a third portion **221c**. First portion **221a** and third portion **221c** may extend approximately parallel to one another, and second portion **221b** may extend between and approximately perpendicular to first portion **221a** and third portion **221c**. Third portion **221c** may extend along protrusion **289** of body **233**. As shown in FIG. 2D, wire **281** may extend from an opening **261** of a block **259** and may enter channel **221** via an opening at a first end **257a** of channel **221**. Wire **281** may extend through channel **221** and exit channel **221** via an opening at a second end **257b** of channel **221**. At least a portion of wire **281** (e.g., a proximal portion) may be welded or otherwise adhered to channel **221**. Third portion **221c** of channel **221** may be configured to align with first channel **219a** of first port **209**. In other words, the opening at second end **257b** of channel **221** may face an opening to channel **219a** of first port **209**. As discussed above, wire **281** may then extend through housing **225** (e.g., through lumen **290**), through first channel **219a**, through the lumen of shaft **252**, and to end effector **256**.

[0044] As shown in FIG. 2D, wire **281** may include a looped or bowed portion **282** or bend to provide strain relief during movement of pivotable member **217** and/or wire **281**. Looped or bowed portion **282** may not be fixed within channel **221**. Alternatively, the looped or bowed portion **282** or bend may be omitted.

[0045] Body **233** of pivotable member **217** may further include an opening **223** for receiving lever **203**. Lever **203** may extend through a first slot **215** formed on an outer surface of housing **225**, through cavity **227**, and through a second slot **255** (see FIG. 2B) within housing **225**. For example, second slot **255** may be positioned below and aligned with first slot **215** and may be formed on an interior surface **257** of housing **225**. First and second slots **215**, **255** may be at a proximal end **297** of housing **225**, and may extend approximately parallel to one another and approximately perpendicular to central longitudinal axis A of first port **209**. In other words, first and second slots **215**, **255** may extend approximately perpendicular to a longitudinal axis of handle **112** when accessory device **201** is coupled to medical device **110**. Slots **215** and **255** may be linear.

[0046] Lever **203** may be pivotable about an axle **247** (FIGS. 2C and 2D) and movable

along/through first and second slots **215**, **255** in a direction R or a direction S (i.e., in directions perpendicular to P/D directions) relative to housing **225**. Slots **215** and **255** may constrain lever **203** to linear motion through slots **215** and **255**. Because lever **203** extends through opening **223**, lever **203** may exert forces on walls of opening **223** to cause pivotable member **217** to move. Movement of lever **203** in direction R may pivot pivotable member **217** in direction G, e.g., such that protrusion **289** moves toward distal end **295** of housing **225**. Movement of lever **203** in direction S may pivot pivotable member **217** in direction H, e.g., such that protrusion **289** moves away from distal end **295** of housing **225**. Opening **223** may have a width that is approximately the same as a width of lever **203** and a length that is greater than a length of lever **203**, such that lever **203** may move along a length of opening **223** as pivotable member **217** pivots.

[0047] In some aspects, wire **281** may be coupled to an electrical socket **229** (shown in FIG. 2B). For example, a proximal end of wire **281** may be received within opening **261** formed on block **259** coupled to electrical socket **229**. Operation portion **241** may be connected to an electrical supply through electrical socket **229** to electrify wire **281**, which may provide, for example, electricity to end effector **256** for electrical cautery or cutting of tissue.

[0048] FIGS. 2A and 2B illustrate accessory device **201** in a first configuration in which lever **203** may be positioned at a first end **251a** of slot **215**. In the first configuration, protrusion **289** of pivotable member **217** may be positioned away from distal end **295** of housing **225**. For example, first edge **245a** of pivotable member **217** may contact or be adjacent to first wall **249a** of cavity **227**. In the first configuration, wire **281** may be in a pulled/tensioned state (i.e., tension may be applied to wire **281**). In other words, wire **281** may be in a proximal configuration. In an example in which end effector **256** is a tome, end effector **256** may be in an actuated state (e.g., a forward bend), as shown in FIG. 3A. In other examples, the first configuration of FIGS. 2A and 2B may correspond to actuated states of other types of end effectors **256** or may correspond to an unactuated state of other types of end effectors **256**.

[0049] Lever **203** may be moved in direction R to transition accessory device **201** to a second configuration in which lever **203** may be positioned at a second end **251b** of slot **215**, as shown in FIGS. 2C and 2D. Movement of lever **203** in direction R along slots **215**, **255** may cause pivotable member **217** to pivot about pivot point **231** in direction G, such that protrusion **289** moves toward distal end **295** of housing **225**. In this configuration, wire **281** may be moved in a distal direction D (e.g., wire **281** may be relaxed). As shown in FIG. 2D, in the second configuration, third portion **221c** of channel **221** may be aligned with lumen **290** and first port **209**. In the second configuration, in an example in which end effector **256** is a tome, end effector **256** may be in an unactuated or relaxed state, as shown in FIG. 3B. In other examples, the second configuration of FIGS. 2C and 2D may correspond to unactuated states of other types of end effectors **256** or may correspond to an actuated state of other types of end effectors **256**.

[0050] To transition accessory device **201** back to the first configuration, lever **203** may be moved in direction S along slots **215**, **255** to pivot pivotable member **217** about pivot point **231** in direction H. Movement of pivotable member **217** in direction H may pull wire **281** proximally (in a proximal direction P), to actuate end effector **256** or return end effector **256** to an unactuated state.

[0051] Alternatively, features of accessory device **201** may be configured such that positioning of lever **203** at an intermediate portion **253** of slot **215** (shown in FIG. 2B) may transition end effector **256** to an unactuated or relaxed state (such as that shown in FIG. 3B for a tome). In such examples, positioning lever **203** at first end **251a** or second end **251b** may cause end effector **256** to transition to a first or a second actuated state. For example, in an example in which end effector **256** is a tome, positioning of lever **203** at second end **251b** of slot **215** may transition end effector **256** to a second actuated state (i.e., a reverse bend), as shown in FIG. 3C.

[0052] During an exemplary medical procedure, an operator may couple accessory device **201** to medical device **110** using strap **205a** and clip **205b**. The operator may then couple syringe **213** to second port **211** of port assembly **207**. The operator may grasp onto handle **112** of medical device

110 with, e.g., their left hand. The operator may navigate shaft **118** of medical device **110** to a position proximate a target site, e.g., a major duodenal papilla. The operator may then insert shaft **252** of accessory device **201** into port **140** of medical device **110** and extend shaft **252** through the working lumen of medical device **110** using, e.g., their right hand. Shaft **252** of accessory device **201**, including end effector **256**, may exit the working lumen of medical device **110**. The operator may use their left hand (or right hand) to control lever **138** to raise elevator **126** and angle end effector **256** toward the papilla. The operator may use their right hand (or left hand) to control lever **203** of accessory device **201** to actuate end effector **256** to cannulate the papilla or perform another medical procedure. The operator may also use their right (or left) hand to operate syringe **213** for delivering fluid during the procedure. Optionally, the operator may rotate port assembly **207** relative to operation portion **241** to move syringe **213** out of the way if, e.g., syringe **213** is impeding the operator's access to lever **203** or to rotate shaft **252**. Thus, a single operator may use medical device **110** and use/actuate accessory device **201**.

[0053] FIG. 4 illustrates an exemplary lever **403** for use with accessory device **201** in place of lever **203**. Lever **403** may include angled portions and/or other ergonomic features to facilitate ease of use. For example, lever **403** may include a first portion **462a**, a second portion **462b**, and a third portion **462c**. First portion **462a** may extend linearly from housing **225** and out of slot **215**, second portion **462b** may extend from and be angled relative to first portion **462a** and toward first end **251a** of slot **215**, and third portion **462c** may extend from and be angled relative to second portion **462b** and toward second end **251b** of slot **215**. Angled portions **462b**, **462c** may allow for an operator to more easily move lever **403** along slot **215** to actuate end effector **256**. In further alternatives, a bent lever or a T-lever may be used in place of levers **203**, **403**. Lever **403** may include a plurality of rounded, protruding portions **486** that may assist an operator with gripping lever **403** or using lever **403** with a single finger.

[0054] FIGS. 5A and 5B illustrate features of an exemplary medical system **500** that includes medical device **110** coupled to an exemplary accessory device **501**, which may have any of the features of accessory device **201**, unless otherwise specified herein. Accessory device **501** may include a port assembly **507** having any of the features of port assembly **207** and an insertion portion **550** having any of the features of insertion portion **250**.

[0055] Accessory device **501** may include an operation portion **541** having a housing **525**. Housing **525** may include or be coupled to one more attachment devices for coupling operation portion **541** to a handle **112** of medical device **110**. For example, housing **525** may be coupled to body **144** of handle **112** via a snap-fit connection. For example, one or more walls of housing **525** may define a recess for receiving body **144** and for securing body **144** with an interference fit. Alternatively, housing **525** may include straps, bands, clips, jaws, or pincers, as discussed above for system **200**.

[0056] Operation portion **541** may include an actuation mechanism **560** for actuating an end effector, e.g., end effector **256**, at a distal end of a shaft **552** of insertion portion **550**. For example, a control wire **581** (shown in FIG. 5B) may extend proximally from end effector **256**, through shaft **552**, through port assembly **507**, and to actuation mechanism **560**.

[0057] As shown particularly in FIG. 5B, actuation mechanism **560** may include an actuator **569**, a first arm **573**, a pin **585**, a second arm **575**, and a connection pin **539** movably coupled to one another. Although the term “connection pin” **539** is used herein, it will be appreciated that connection pin **539** may include a bar, a rod, or any other suitable structure. Actuator **569** may be positioned at a proximal portion **565** of housing **525**.

[0058] Actuator **569** may be, e.g., a toggle or a lever, and may include a first contact element **571a** (e.g., a contact surface) and a second contact element **571b** (e.g., a contact surface). First and second contact elements **571a**, **571b** may include concave surfaces for ease of use. For example, an operator may apply a force on first contact element **571a** to move first contact element **571a** toward body **144** and second contact element **571b** away from body **144**. Conversely, an operator may apply a force on second contact element **571b** to move second contact element **571b** toward body

144 and first contact element **571a** away from body **144**. Alternatively, actuator **569** may be or may include a switch, button, knob, a lever, and/or other structure.

[0059] As shown in FIG. 5B, actuator **569** may be coupled to a first end **579a** of first arm **573** via e.g., a hinge **587**. In examples, first end **579a** may be pivotably/rotationally coupled to actuator **569**. First arm **573** may extend distally within housing **525** to second arm **575** within and at a distal portion **567** of housing **525**. A second end **579b** of first arm **573** may be coupled to a first end **583a** of second arm **575**. First arm **573** may be movable in generally proximal and distal directions (e.g., in an axial direction), and second arm **575** may be pivotable about pin **585**. A second end **583b** of second arm **575** may be pivotably/rotationally coupled to a connection pin **539**.

[0060] Connection pin **539** may be movable along a channel **577** formed at distal portion **567** of housing **525**. Channel **577** may extend approximately parallel to a longitudinal axis of housing **525**, such that connection pin **539** may move in proximal/distal directions along channel **577**.

Connection pin **539** may be coupled to a proximal end of wire **581**, such that movement of connection pin **539** in proximal and distal directions may cause movement of wire **581** in the corresponding proximal and distal directions to actuate end effector **256**. Connection pin **539** may also be configured to couple to an electrical supply to electrify wire **581**.

[0061] FIGS. 5A and 5B illustrate a first configuration of accessory device **501** in which wire **581** is in a distal configuration (e.g., an actuated configuration of the tome shown in FIGS. 3A-3C). In use, an operator may grasp onto, e.g., a proximal portion of handle **112** using their left hand. The operator may then place, e.g., their ring finger on first contact element **571a** and their pinky finger on second contact element **571b**. The operator may apply a force onto first contact element **571a** using their ring finger to move first contact element **571a** toward body **144** of handle **112** transitioning accessory device **501** to a second configuration. Movement of first contact element **571a** toward body **144** of handle **112** may cause first arm **573** to move distally. Distal movement of first arm **573** may cause second arm **575** to pivot about pin **585**, e.g., counterclockwise in the frame of reference shown in FIG. 5B. Such movement of second arm **575** may cause connection pin **539** to move proximally along channel **577**, which in turn may move or pull wire **581** proximally to actuate end effector **256** (or to transition end effector **256** to an unactuated state for a device that is actuated by moving pull wire **581** distally).

[0062] The operator may apply a force onto second contact element **571b** using their pinky finger to transition accessory device **501** back to the first configuration. Movement of second contact element **571b** toward body **144** of handle **112** may cause first arm **573** to move proximally. Proximal movement of first arm **573** may cause second arm **575** to pivot about pin **585** in an opposite direction to that described above, e.g., clockwise in the frame of reference shown in FIG. 5B. Such movement of second arm **575** may cause connection pin **539** to move distally along channel **577**, which in turn may move or push wire **581** distally to transition end effector **256** to an unactuated or relaxed state (or to an actuated state of a medical device that is unactuated by moving wire **581** proximally).

[0063] FIGS. 6A and 6B illustrate internal and external features of an exemplary medical system **600** that includes medical device **110** coupled to an exemplary accessory device **601**. Accessory device **601** may have any of the features of accessory devices **201**, **501**, unless otherwise specified herein. FIG. 6A illustrates a front view of medical system **600** and FIG. 6B illustrates a back view of medical system **600**. Accessory device **601** may include a port assembly **607** having any of the features of port assemblies **207**, **507**. Port assembly **607** may be configured to couple to an insertion portion (not shown) having any of the features of insertion portions **250**, **550**. For example, the insertion portion of accessory device **601** may include a shaft (not shown) extending distally from port assembly **607** and a distal end of the shaft may include an end effector (not shown), e.g., end effector **256**. A control wire (not shown for clarity of illustration in FIGS. 6A and 6B but having any of the properties of control wires **281**, **581** may extend proximally from end effector **256** (or another end effector), through the shaft, through port assembly **607**, and to an

operation portion **641** of accessory device **601**. The control wire may be coupled, directly or indirectly, to a first actuation mechanism **660** and a second actuation mechanism **696** of operation portion **641** for actuating end effector **256** (or an alternative end effector), which will be discussed in greater detail below.

[0064] Operation portion **641** may include a housing **625** having one more attachment devices for coupling operation portion **641** to handle **112** of medical device **110**. For example, housing **625** may be coupled to body **144** of handle **112** via a snap-fit connection. As shown in FIG. **6A**, one or more walls of housing **625** may define a recess **698** on a front side of housing **625** for receiving body **144** and for securing body **144** with an interference fit. Additionally or alternatively, housing **625** may include straps, bands, clips, jaws, or pincers, as discussed above for system **200**, to secure operation portion **641** to handle **112**.

[0065] As mentioned above, operation portion **641** may include first actuation mechanism **660** and second actuation mechanism **696** coupled to the control wire. First actuation mechanism **660** may be configured to move the control wire proximally and distally, and second actuation mechanism **696** may be configured to rotate the control wire. The control wire or portions of the control wire (e.g., proximal portions) may have any noncircular shape (e.g., a ribbon shape, a rectangle shape, etc.) to rotatably fix (e.g., key) the control wire to a second wheel **672b** (FIG. **6B**) of second actuation mechanism **696**, which will be discussed in greater detail below. In some examples, only a portion of the control wire that extends through second wheel **672b** may have a non-round shape.

[0066] First actuation mechanism **660** may include a slider **635** (e.g., a first actuator) movable proximally and distally along a slot formed on an outer surface on a first lateral side **625a** of housing **625**. A proximal portion **664** of slider **635** may include a contact element **671** (e.g., a body) having an outer, concave surface **671a** extending between first and second outer, planar surfaces **671b**, **671c** (labeled in FIG. **6B**). Concave surface **671a** may face away from housing **625**, and first and second planar surfaces **671b**, **671c** may extend perpendicular to a longitudinal axis of housing **625**.

[0067] As more clearly shown in FIG. **6B**, a distal portion **668** of slider **635** may include a first wall **691** recessed from contact element **671** and extending parallel to a longitudinal axis of housing **625**. Outer, planar surface **671b** of contact element **671** and an outer, planar surface **691a** of first wall **691** may define a corner **693** of slider **635**. An inner surface of contact element **671** and an inner surface of first wall **691** may form a continuous first inner surface of slider **635** extending parallel to a longitudinal axis of housing **625** and slidable along the outer surface on first lateral side **625a** of housing **625** and/or the slot of housing **625**. The shape of slider **635** discussed above may allow an operator to actuate slider **635** without looking at slider **635** and/or retain a finger of the operator along concave surface **671a**. Additionally or alternatively, an operator may rest his or her finger on planar surface **671b** when operating slider **635**. The shape of slider **635** may thus offer multiple ergonomic options for an operator. However, the shape of slider **635** described above is merely exemplary.

[0068] As shown in FIG. **6A**, slider **635** may include a second wall **692** disposed on a front side of housing **625**. Second wall **692** may extend from a proximalmost end of slider **635** to a distalmost end of slider **635**, and may extend approximately perpendicularly from contact element **671** and first wall **691**. Second wall **692** may define a second inner surface of slider **635** slidable along an outer surface on the back side of housing **625** (e.g., an outer surface that extends approximately perpendicularly from the outer surface on first side **625a** of housing **625**).

[0069] Alternatively, second wall **692** may be omitted from slider **635**. As shown in FIG. **6B**, slider **635** may further include a projection **602** coupled to the inner surface of contact element **671**. Projection **602** may be movable within housing **625** and along the slot of housing **625**. Projection **602** may be coupled to and/or may extend from a portion of the inner surface of contact element **671** adjacent to the proximalmost end of slider **635**. Alternatively, projection **602** may be coupled to and/or may extend from any portion of the inner surface of contact element **671** and/or first wall

691. The control wire may be coupled to or fixed to protrusion **602**, such that movement of slider **635** proximally or distally along the slot of housing **625** may move the control wire proximally or distally.

[0070] Second actuation mechanism **696** may include a knob **684** (e.g., a second actuator) disposed on a second lateral side **625b** of housing **625** and at a proximal end **697** of housing **625**. For example, knob **684** may be disposed within a recess of housing **625**. Knob **684** may include a plurality of indentations **673** or other gripping features that may assist an operator with rotating knob **684** with, e.g., their thumb. As shown in FIG. **6B**, second actuation mechanism **696** may further include a rod **643**, and first and second wheels **672a**, **672b** disposed within housing **625** and on opposite sides of housing **625**. For example, first wheel **672a** may be positioned closer to second lateral side **625b** and second wheel **672b** may be positioned closer to first lateral side **625a**. A diameter of first wheel **672a** may be greater than a diameter of second wheel **672b**. Alternatively, a diameter of first wheel **672a** may have a same diameter as second wheel **672b** or second wheel **672b** may have a greater diameter.

[0071] Portions of rod **643** may extend through knob **684** and remaining portions of rod **643** may extend within housing **625**. Rod **643** may extend distally through knob **684** and towards a central longitudinal axis of housing **625** to first wheel **672a**. First and second ends of rod **643** may be fixed to knob **684** and to first wheel **672a**, respectively. Thus, rotation of knob **684** about a rotation axis Y in a direction (e.g., clockwise or counterclockwise) may rotate rod **643** and first wheel **672a** about rotation axis Y in the same direction. Axis Y may be transverse (not parallel to) the central longitudinal axis of housing **625**.

[0072] Second wheel **672b** may be distal to and aligned with protrusion **602** of slider **635**, and substantially aligned with first wheel **672a** along an axis extending approximately perpendicularly to a longitudinal axis of housing **625**. First wheel **672a** and second wheel **672b** may be coupled to one another via, e.g., a belt or chain (not shown but having features of a belt **878**, discussed below) extending from first wheel **672a** to second wheel **672b**, such that rotation of first wheel **672a** about rotation axis Y in a direction (e.g., clockwise or counterclockwise) rotates second wheel **672b** about a rotation axis X. As shown in FIG. **6B**, rotational axis Y may extend transversely (not parallel) to rotational axis X. In other words, axis Y may have a non-zero angle with respect to axis X. Alternatively, knob **684**, rod **643**, and wheel **672a** may be arranged such that rotational axis Y may extend parallel to or approximately parallel to rotational axis X.

[0073] The control wire may extend distally from protrusion **602** and through an opening of second wheel **672b** (e.g., an opening having any of the properties of opening **774** below). As mentioned above, the control wire (or portions of the control wire) may have any noncircular shape. A size and a shape of the opening of second wheel **672b** may be keyed to/complementary to a size and shape of the control wire, which may allow for the control wire to move proximally or distally through the opening of second wheel **672b**, but may prevent rotation of the control wire relative to second wheel **672b**. In other words, the control wire may be freely movable proximally or distally (e.g., in a longitudinal direction) within the opening of second wheel **672b** and relative to second wheel **672b** due to proximal/distal movement of slider **635**. However, the control wire may be rotatably fixed to second wheel **672b**. For example, rotation of knob **684** and thus rotation of second wheel **672b** about rotation axis X in a direction (e.g., clockwise or counterclockwise), rotates the control wire about rotation axis X in the same direction.

[0074] FIGS. **6A** and **6B** illustrate slider **635** in a proximal position in which the control wire may be in a pulled/tensioned state (e.g., a proximal configuration). In an example in which end effector **256** is a tome, end effector **256** may be in an actuated state (e.g., a forward bend), as shown in FIG. **3A**, when slider **635** is in the proximal position. To transition the control wire to a relaxed state (e.g., a distal configuration), an operator may place their ring finger or other finger on concave surface **671a** of contact element **671** and push downward/distally on contact element **671** to move slider **635** distally along the slot of housing **625** to a distal position. In an example in which end

effector **256** is a tome, end effector **256** may be in an unactuated or relaxed state, as shown in FIG. 3B, when slider **635** is in the distal position. The operator may use their thumb of the same hand to rotate knob **684** to rotate the control wire/end effector **256**.

[0075] FIGS. 7A and 7B illustrate features of an exemplary medical system **700** that includes medical device **110** coupled to an exemplary accessory device **701**. Accessory device **701** may have any of the features of accessory devices **201**, **501**, **601**, unless otherwise specified herein. FIG. 7A illustrates a cross-sectional view of a portion of medical system **700**. Accessory device **701** may include a port assembly **707** having any of the features of port assemblies **207**, **507**, **607**. Port assembly **707** may be configured to couple to an insertion portion (not shown) having any of the features of insertion portions **250**, **550**. For example, the insertion portion of accessory device **701** may include a shaft (not shown) extending distally from port assembly **707** and a distal end of the shaft may include an end effector (not shown), e.g., end effector **256**. A control wire (not shown but having the features of any of control wires **281**, **581**) may extend proximally from end effector **256**, through the shaft, through port assembly **707**, and to an operation portion **741** of accessory device **701**. The control wire may be coupled to a first actuation mechanism (not shown in FIGS. 7A and 7B) of operation portion **741** having any of the features of actuation mechanisms **260**, **560**, **660** for controlling proximal and distal movement of the control wire to control aspects of end effector **256**, as described above.

[0076] The control wire may be further coupled to a second actuation mechanism **796** of operation portion **741**. Second actuation mechanism **796** may be configured to rotate the control wire and thus rotate end effector **256**. Second actuation mechanism **796** may include a knob **784** disposed on a side of a housing **725** of operation portion **741**. A plurality of indentations **773** or other gripping features may be disposed around a circumference/perimeter of knob **784** to assist an operator with rotating knob **784**. Knob **784** may function as a gear **704**. An outer side of knob **784** may include a planar surface **784a** and an inner side of knob **784** may form first gear **704** (shown in FIG. 7B). For example, knob **784**/gear **704** may form a crown gear, having teeth **778** facing/extending inward (i.e., a direction approximately perpendicular to planar surface **784a**, toward a central longitudinal axis of device **110**). Teeth **778** may be directly molded onto a surface on the inner side of knob **784**.

[0077] As more clearly shown in FIG. 7B, teeth **778** of first gear **704** may be configured to engage with teeth **780** of a second gear **706**, e.g., a pinion gear, disposed within housing **725**. Engagement of teeth **778** and teeth **780** may be configured to rotate second gear **706** upon rotation of knob **784**. Second gear **706** may be oriented approximately perpendicular to first gear **704**, such that a rotational axis D of second gear **706** may be approximately perpendicular to a rotational axis E of knob **784**/first gear **704**. Although FIG. 7B does not show all of the features of knob **784** (e.g., indentations **773**) for ease of illustration, it will be appreciated that any of such features may be disposed on knob **784** of FIG. 7B.

[0078] Similar to the control wire of FIGS. 6A and 6B, the control wire of FIGS. 7A and 7B or a portion thereof may also have any noncircular shape. The control wire may extend through a central opening **774** of second gear **706**. A size and a shape of central opening **774** of second gear **706** may be keyed to/complementary to a size and shape of the control wire, which may allow for the control wire to move proximally or distally through central opening **774** of second gear **706**, but may prevent rotation of the control wire relative to second gear **706**. For example, the control wire and central opening **774** of second gear **706** may each have a rectangle shape; however, such a shape is merely exemplary and any suitable non-round shape may be utilized (e.g., oval, elongate, polygonal, etc.). The control wire may be rotatably fixed relative to second gear **706** such that rotation of second gear in a direction (e.g., clockwise or counterclockwise) about rotational axis D rotates the control wire in the same direction about rotational axis D.

[0079] In use, an operator may, e.g., rotate knob **784** clockwise about rotational axis E, which in turn rotates second gear **706** and the control wire clockwise about rotational axis D, to rotate end effector **256** to a first position. Axis E may be approximately perpendicular to a central longitudinal

axis of medical device **110**, and rotational axis **D** may be approximately parallel to the central longitudinal axis of medical device **110**. The operator may then, e.g., rotate knob **784** counterclockwise about rotational axis **E**, which in turn rotates second gear **706** and the control wire counterclockwise about rotational axis **D**, to rotate end effector **256** to a second position. [0080] FIGS. **8A-8C** illustrate internal and external features of an exemplary medical system **800** that includes medical device **110** coupled to an exemplary accessory device **801**. Accessory device **801** may have any of the features of accessory devices **201**, **501**, **601**, **701**, unless otherwise specified herein. Accessory device **801** may include a port assembly **807** having any of the features of port assemblies **207**, **507**, **607**, **707**. Port assembly **807** may be configured to couple to an insertion portion (not shown) having any of the features of insertion portions **250**, **550**. For example, the insertion portion of accessory device **801** may include a shaft (not shown) extending distally from port assembly **807** and a distal end of the shaft may include an end effector (not shown), e.g., end effector **256**. A control wire (not shown but having any of the features of control wires **281**, **581**) may extend proximally from end effector **256**, through the shaft, through port assembly **807**, and to an operation portion **841** of accessory device **801**.

[0081] The control wire may be coupled (directly or indirectly) to a first actuation mechanism **860** and a second actuation mechanism **896** of operation portion **841** for actuating end effector **256**, which will be discussed in greater detail below. Operation portion **841** may include a housing **825** having one more attachment devices for coupling operation portion **841** to handle **112** of medical device **110**. For example, operation portion **841** may include a flexible clip **805b** similar to clip **205b** configured to clip onto sides of port **140**. Additionally or alternatively, operation portion **841** may include straps, bands, jaws, or pincers, as discussed above for system **200**, to secure operation portion **841** to handle **112**.

[0082] As mentioned above, operation portion **841** may include first actuation mechanism **860** and second actuation mechanism **896** coupled to the control wire to control aspects of end effector **256**. First actuation mechanism **860** may be configured to move the control wire proximally and distally, and second actuation mechanism **896** may be configured to rotate the control wire. Similar to the control wires of FIGS. **6A-6B** and **7A-7B**, the control wire of FIGS. **8A-8C** may also have any noncircular shape.

[0083] Actuation mechanism **860** may include a lever **803** (shown in FIG. **8B**), e.g., a first actuator, that may be operable to pivot or rotate a pivotable member **817**. Lever **803** may have a substantially straight shape and may be disposed on a first side **825a** of housing **825**. The control wire may be coupled to pivotable member **817** such that movement of pivotable member **817** may cause movement of the control wire. Pivotable member **817** may include a body **833** positioned within a cavity **827** of housing **825**. Body **833** may have a substantially L shape. For example, body **833** may include a first arm **833a** and a second arm **833b** extending away from a pivot point **831**, and a length of first arm **833a** may be greater than a length of second arm **833b**. Body **833** may further include an opening **823** on second arm **833b** configured to receive lever **803** and a protrusion **889** at an end of first arm **833a**. Body **833** may be pivotable about pivot point **831**. Pivot point **831** may include an axle, such as a pin, rivet, or similar structure that may be received within an opening of body **833** (e.g., an opening positioned at a vertex of body **833**), such that pivotable member **817** is rotationally/pivotably movable about pivot point **831**.

[0084] Lever **803** may be pivotable about an axle (not shown) and movable along/through a slot (not shown) formed on an outer surface of housing **825**. For example, the slot may be similar to slot **215** of FIGS. **2A-2D**, except positioned closer to first side **825a** of housing **825**, and the slot of device **801** may be curved. Lever **803** may be moved (e.g., along an arcuate or straight path) to move the control wire proximally or distally. Movement of lever **803** in a first direction may pivot pivotable member **817** about pivot point **831** such that first arm **833a** may move away from a distal end **895** of housing **825**, and thus pulling the control wire proximally. Movement of lever **803** in a second direction may pivot pivotable member **817** about pivot point **831** such that first arm **833a**

may move towards distal end **895** of housing **825**, and thus moving the control wire distally. [0085] As mentioned above, operation portion **841** may include second actuation mechanism **896** for rotating the control wire. Second actuation mechanism **896** may include a knob **884** (e.g., a second actuator) disposed on first side **825a** of housing **825** and distal to lever **803**. Knob **884** may be oriented on/within housing **825**, such that a rotational axis M of knob **884** may be approximately parallel to a longitudinal axis of housing **825**. An entire circumference of knob **884** may include a plurality of indentations **873** or other gripping features that may assist an operator with rotating knob **884**.

[0086] Second actuation mechanism **896** may further include first and second wheels **872a**, **872b** disposed within housing **825** and on opposite sides of housing **825**. First wheel **872a** may be rotatably fixed to knob **884**. For example, first wheel **872a** may be attached to a proximal wall/surface of knob **884**. Rotation of knob **884** about rotation axis M in a direction (e.g., clockwise or counterclockwise) may rotate first wheel **872a** about rotation axis M in the same direction. Second wheel **872b** may be aligned with first wheel **872a** such that a rotational axis N of second wheel **872b** may be parallel to rotational axis M of knob **884**/first wheel **872a**. Second wheel **872b** may also be aligned axially with first wheel **872a**. As shown in FIGS. **8B** and **8C**, first wheel **872a** and second wheel **872b** may be coupled to one another via belt **878** extending from first wheel **872a** to second wheel **872b**, such that rotation of knob **884**/first wheel **872a** about rotation axis M in a direction (e.g., clockwise or counterclockwise) rotates second wheel **872b** about rotation axis N in the same direction. A diameter of first wheel **872a** may be greater than a diameter of second wheel **872b**. Alternatively, a diameter of first wheel **872a** may have a same diameter as second wheel **872b**, or second wheel **872b** may have a greater diameter than first wheel **872a**.

[0087] FIG. **8C** illustrates a top view of features of actuation mechanism **896** (e.g., a top portion of housing **825** removed to expose features of actuation mechanism **896**). The control wire may extend distally from pivotable member **817** and through a central opening **870** of second wheel **872b**. As mentioned above, the control wire may have any noncircular shape. A size and a shape of central opening **870** of second wheel **872b** may be keyed to/complementary to a size and shape of the control wire, which may allow for the control wire to move proximally or distally through central opening **870** of second wheel **872b**, but may inhibit rotation of the control wire relative to second wheel **872b**. For example, rotation of second wheel **872b** about rotation axis N in a direction (e.g., clockwise or counterclockwise) rotates the control wire about rotation axis N in the same direction.

[0088] During an exemplary medical procedure, an operator may rotate knob **864**/first wheel **872a** clockwise about rotational axis M, which in turn rotates second wheel **872b** and the control wire clockwise about rotational axis N. Rotation of second wheel **872b** and the control wire clockwise about rotational axis N may rotate end effector **256** to a first position. An operator may then rotate knob **864**/first wheel **872a** counterclockwise about rotational axis M, which in turn rotates second wheel **872b** and the control wire counterclockwise about rotational axis N. Rotation of second wheel **872b** and the control wire counterclockwise about rotational axis N may rotate end effector **256** to a second position. To transition end effector **256** between actuated and unactuated states, the operator may move lever **803** to move the control wire proximally or distally.

[0089] FIG. **9** illustrates features of an exemplary second actuation mechanism **996** that may be used with accessory device **801**, e.g., in place of second actuation mechanism **896**. Second actuation mechanism **996** may also be used with other aspects disclosed herein. For example, the control wire may be coupled to first actuation mechanism **860** for controlling proximal/distal movement of the control wire and second actuation mechanism **996** for controlling rotation of the control wire. Actuation mechanism **996** may include a gear assembly **999** coupled to a knob (not shown) similar to knob **884**. In this example, the knob and gear assembly **999** may be positioned on a second side **825b** of housing **825**, opposite first side **825a**. Gear assembly **999** may include a first, outer gear **904** having teeth **978** (e.g., internal teeth) extending radially inward towards a rotational

axis of outer gear **904**. Outer gear **904** may be rotatably fixed to the knob. Gear assembly **999** may further include a second, inner gear **906** having teeth **980** (e.g., external teeth) extending radially outward, away from a rotational axis of inner gear **906**. Teeth **978** of outer gear **904** may be configured to engage with teeth **980** of inner gear **906** to rotate inner gear **906**. Inner gear **906** may include a central opening **974** having a noncircular shape (e.g., a rectangle shape) for receiving the control wire having a complementary, noncircular shape. As such, the control wire may be rotatably fixed to inner gear **906**. Rotation of the knob/outer gear **904** in a direction (e.g., clockwise or counterclockwise) rotates inner gear **906** and the control wire/end effector **256** in a same direction.

[0090] FIG. **10** illustrates an exemplary medical system **1000** that includes medical device **110** coupled to an exemplary accessory device **1001**. Accessory device **1001** may have any of the features of accessory devices **201**, **501**, **601**, **701**, **801**, unless otherwise specified herein. Accessory device **1001** may include a port assembly (not shown) having any of the features of port assemblies **207**, **507**, **607**, **707**, **807**. The port assembly may be configured to couple to an insertion portion (not shown) having any of the features of insertion portions **250**, **550**. For example, the insertion portion of accessory device **1001** may include a shaft (not shown) extending distally from the port assembly and a distal end of the shaft may include an end effector (not shown), e.g., end effector **256**. A control wire (not shown but having any of the features of control wires **281**, **581**) may extend proximally from end effector **256**, through the shaft, through port assembly, and to an operation portion **1041** of accessory device **1001**. The control wire may be coupled to a first actuation mechanism **1060** and a second actuation mechanism **1096**. Second actuation mechanism **1096** may have any of the features of actuation mechanism **896**. For example, second actuation mechanism **1096** may include a knob **1084** having any of the features of knob **884** for controlling rotation of the control wire/end effector **256**.

[0091] First actuation mechanism **1060** may be configured to control proximal/distal movement of the control wire to transition end effector **256** between actuated and unactuated states. First actuation mechanism **1060** may include a slider **1035** movable along a slot **1015** formed on a front side of housing **1025**. Slot **1015** may be a linear slot extending approximately parallel to a longitudinal axis of housing **1025** and/or handle **112**. Slider **1035** may include a contact element **1071** similar to contact element **671** at a proximal portion of slider **1035**. Contact element **1071** may include a concave surface **1071a**.

[0092] An optional post **1094** may extend from a bottom of concave surface **1071a** and approximately perpendicular to a longitudinal axis of housing **1025**. Post **1094** may assist an operator with moving slider **1035** along slot **1015**. For example, an operator may push downward on post **1094** to move slider distally along slot **1015** to move the control wire distally. For example, the operator may push upward on post **1094** to move slider proximally along slot **1015** to move the control wire proximally. In some examples, post **1094** may be integral with remaining portions of slider **1035**. In other examples, post **1094** may be a detachable/attachable piece that may be received within an opening formed on concave surface **1071a**.

[0093] FIG. **11** illustrates an exemplary medical system **2000** that includes an exemplary accessory device **2001** coupled to medical device **110**. Accessory device **2001** may include one or more attachment devices for coupling accessory device **2001** to handle **112** of medical device **110**. For example, accessory device **2001** may include a clip **2005** configured to couple to handle **112** of medical device **110**. Additionally or alternatively, accessory device **2001** may include straps, bands, jaws, and/or pincers, as discussed above for system **200**, to secure accessory device **2001** to handle **112** of medical device **110**.

[0094] Accessory device **2001** may include an operation portion **2041** coupled to a port assembly **2007** having any of the features of port assemblies **207**, **507**, **607**, **707**, **807**. For example, port assembly **2007** may include a first port **2009** configured to couple to a main body **2100** of operation portion **2041** such that first port **2009** may be in fluid communication with one or more cavities

and/or slots of main body **2100**. First port **2009** may secure port assembly **2007** to operation portion **2041**, while allowing main body **2100** to rotate relative to first port **2009**. Port assembly **2007** may include a second port **2011** configured to couple to a syringe **213** or other fluid source for delivering fluid to a target site. Although port assembly **2007** is illustrated as having a Y-shape, port assembly **2007** may alternatively have a T-shape.

[0095] Port assembly **2007** may be configured to couple an insertion portion (not shown) having any of the features of insertion portions **250**, **550**. For example, the insertion portion of accessory device **2001** may include a shaft (not shown) extending distally from port assembly **2007** and a distal end of the shaft may include an end effector (not shown), e.g., end effector **256**. A control wire (not shown but having any of the properties of control wires **281**, **581**) may extend proximally from end effector **256**, through the shaft, through first port **2009**, and into a cavity and/or slot **2114** of main body **2100**.

[0096] A proximal end of the control wire may be coupled to an internal connector of actuator **2102** (e.g., a knob, a spool) of operation portion **2041**, e.g., an internal connector within second flange portion **2104b** of actuator **2102**, to secure the control wire and actuator **2102** to main body **2100**. Actuator **2102** may be configured to move in the proximal/distal directions (e.g., an axial direction) along slot **2114** to move the control wire in the proximal/distal directions. As mentioned above, main body **2100** may be configured to rotate relative to first port **2009**. Because the control wire is secured to main body **2100**, rotation of main body **2100** relative to first port **2009** rotates the control wire. Actuator **2102** may rotate with main body **2100** and thus may be used to rotate the control wire.

[0097] Actuator **2102** may include a first flange portion **2104a**, a second flange portion **2104b**, and a central portion **2108** (e.g., an indented portion) extending between first and second flange portions **2104a**, **2104b**. A diameter of each of first and second flange portions **2104a**, **2104b** may be greater than a diameter of central portion **2108**. First flange portion **2104a** may include a plurality of indentations **2073** around an entire circumference of flange portion **2104a**. Indentations **2073** may assist an operator with gripping first flange portion **2104a** to rotate actuator **2102**/main body **2100** and/or serve as markers to indicate to an operator that actuator **2102** is rotatable. Additionally or alternatively, first flange portion **2104a** may include any combination of gripping features and/or rotational markers, e.g., bumps, flat surfaces, recesses, etc.

[0098] Actuator **2102** may include a connection portion **2110** having a plug coupled to the control wire. The plug may be configured to couple to an electrical supply to electrify the control wire, which may provide, for example, electricity to end effector **256** for electrical cautery or cutting of tissue. As illustrated in FIG. **11**, connection portion **2110** may extend from second flange portion **2104b** perpendicularly to a central longitudinal axis of actuator **2102**. Alternatively, connection portion **2110** may extend from second flange portion **2104b** transversely to a central longitudinal axis of actuator **2102**. In another alternative, connection portion **2110** may be omitted from actuator **2102** and/or may be disposed on any portion of main body **2100**.

[0099] Operation portion **2041** may further include a proximal body **2112** coupled to a proximal end of main body **2100**. Proximal body **2112** may be fixed to main body **2100** such that proximal body **2112** and main body **2100** may rotate together. In some aspects, as illustrated in FIG. **11**, proximal body **2112** may serve as a thumb rest when an operator is using other finger(s) to move actuator **2102** proximally or distally along slot **2114**. Additionally or alternatively, an operator may place their thumb and their index finger on proximal body **2112** to rotate proximal body **2112**/main body **2100** relative to first port **2009** and thus rotate the control wire/end effector **256**. Similar to actuator **2102**, proximal body **2112** may include any combination of gripping features and/or rotational markers, e.g., bumps, flat surfaces, recesses, etc.

[0100] In use, an operator may place central portion **2108** of actuator **2102** in between their index finger and middle finger, and may place their thumb on proximal body **2112**, as shown in FIG. **11**. The operator may use their index finger and middle finger to move actuator **2102** proximally along

slot **2114** to transition the control wire to a pulled/tensioned state (e.g., a proximal configuration). In the proximal configuration, in an example in which end effector **256** is a tome, end effector **256** may be in an actuated state (e.g., a forward bend), as shown in FIG. **3A**. In other examples, the proximal configuration may correspond to actuated states of other types of end effectors **256** or may correspond to an unactuated state of other types of end effectors **256**.

[0101] To transition the control wire to a relaxed state (e.g., a distal configuration), the operator may move actuator **2102** distally along slot **2114**. In the distal configuration, in an example in which end effector **256** is a tome, end effector **256** may be in an unactuated or relaxed state, as shown in FIG. **3B**. In other examples, the second configuration may correspond to unactuated states of other types of end effectors **256** or may correspond to an actuated state of other types of end effectors **256**. To control rotation of end effector **256**, the operator may place their thumb and index finger on first flange portion **2104a** to rotate actuator **2102**/main body **2100**/proximal body **2112** relative to first port **2009** or the operator may place their thumb and index finger on proximal body **2112** to rotate actuator **2102**/main body **2100**/proximal body **2112** relative to first port **2009**.

[0102] FIG. **12** illustrates an exemplary accessory device **3001** configured to couple to medical device **110**. Accessory device **3001** may have any of the features of accessory device **2001**, unless otherwise specified herein. Accessory device **3001** may include an attachment device **3005**, a port assembly **3007**, an operation portion **3041** having a body **3100**, and an insertion portion (not shown) having any of the features of insertion portions **250**, **550**. Accessory device **3001** may optionally be coupled to syringe **213** or other fluid source. Similar to accessory device **2001**, a control wire (not shown but having any of the features of control wires **281**, **581**) may extend proximally from end effector **256** and to an internal connector of an actuator **3102** having any of the features of actuator **2102**. Actuator **3102** may be configured to move in the proximal/distal directions along a slot of main body **3100** similar to slot **2114** of main body **2100** to move the control wire in the proximal/distal directions to transition end effector **256** between actuated and unactuated states. Actuator **3102** may be further configured to rotate with main body **3100** relative to a first port **3009** of port assembly **3007** to rotate the control wire/end effector **256**. Actuator **3102** may optionally include a connection portion **3110** similar to connection portion **2110**.

[0103] In this example, actuator **3102** may include a contact element **3116** at a proximal portion of actuator **3102**. Contact element **3116** may include alternating recesses **3118** and ridges **3120**. For example, contact element **3116** may be substantially star-shaped having pointed recesses **3118**/ridges **3120**. In other examples, contact element **3116** may have curved recesses **3118**/ridges **3120**. Contact element **3116** may include any number of recesses **3118** and ridges **3120**.

[0104] FIGS. **13-15** illustrate exemplary actuators **4102**, **5102**, **6102** that may be used with main bodies **2100**, **3100** of accessory devices **2001**, **3001** (e.g., in place of actuators **2102**, **3102**). However, it will be appreciated that the features of actuators **2102**, **3102**, **4102**, **5102**, **6102** described herein may be combined in any way to e.g., improve actuator ergonomics and usability, serve as markers to an operator that the actuator is rotatable in addition to being movable in the proximal/distal directions, and/or serve as rotation markers, e.g., to assist an operator track how much main bodies **2100**, **3100** have been rotated.

[0105] FIG. **13** illustrates exemplary actuator **4102** having any of the features of actuators **2102**, **3102**, unless otherwise specified herein. Actuator **4102** may include a first flange portion **4104a**, a second flange portion **4104b**, and a central portion **4108** (e.g., an indented portion) extending between first and second flange portions **4104a**, **4104b**. Actuator **4102** may include a connection portion **4110** similar to connection portions **2110**, **3110**. Actuator **4102** may include an opening **4122** for receiving main bodies **2100**, **3100**.

[0106] Central portion **4108** and second flange portion **4104b** may include smooth, outer surfaces. First flange portion **4104a** may include alternating protrusions **4132** and indentations **4130** disposed around an entire circumference of first flange portion **4104a**. Each protrusions **4132** and each indentation **4130** may extend parallel to a central longitudinal axis of actuator **4102**. An outer

surface of each indentation **4130** may be substantially flat or substantially pointed. Although second flange portion **4104b** is illustrated as having a smooth surface, second flange portion **4104b** may alternatively include alternating protrusions **4132** and indentations **4130**.

[0107] FIG. **14** illustrates exemplary actuator **5102** having any of the features of actuators **2102**, **3102**, **4102**, unless otherwise specified herein. Actuator **5102** may include a first flange portion **5104a**, a second flange portion **5104b**, and a central portion **5108** (e.g., an indented portion) extending between first and second flange portions **5104a**, **5104b**. Actuator **5102** may include an opening **5122** for receiving main bodies **2100**, **3100**.

[0108] A connection portion, such as connection portions **2110**, **4110**, may be omitted or may be present (but not shown). Central portion **5108** and second flange portion **5104b** may include smooth, outer surfaces. First flange portion **5104a** may include one or more flat sections **5124** and one or more curved sections **5126**. As illustrated in FIG. **14**, first flange portion **5104a** may include two flat sections **5124** disposed on opposite sides of actuator **5102** and two curved sections **5126** disposed on opposite sides of actuator **5102**. Each flat section **5124** may include a planar surface and each curved section **5126** may include a curved surface. The curved surface of curved section **5126** may include a plurality of protrusions **5132** extending radially outward from the curved surface. Protrusions **5132** may extend between a distalmost edge of first flange portion **5104a** and a proximal, rounded edge of first flange portion **5104a**. For example, protrusions **5132** may terminate distal to the proximal, rounded edge of first flange portion **5104a**. Although second flange portion **5104b** is illustrated as having a smooth surface, second flange portion **5104b** may alternatively include one or more flat sections **5124** and/or one or more curved sections **5126**.

[0109] FIG. **15** illustrates exemplary actuator **6102** having any of the features of actuators **2102**, **3102**, **4102**, **5102**, unless otherwise specified herein. Actuator **6102** may include a first flange portion **6104a**, a second flange portion **6104b**, and a central portion **6108** (e.g., an indented portion) extending between first and second flange portions **6104a**, **6104b**. Actuator **6102** may include an opening **6122** for receiving main bodies **2100**, **3100**.

[0110] A connection portion, such as connection portions **2110**, **4110**, may be omitted or may be present (but not shown). Central portion **6108** and second flange portion **6104b** may include smooth, outer surfaces. First flange portion **6104a** may include alternating sets of thin, linear protrusions **6132a** and protrusions **6132b** having an elongated, dome shape. For example, a width of protrusion **6132b** may be greater than a width of protrusion **6132a**. Protrusions **6132a**, **6132b** may extend radially outward from an outer surface of first flange portion **6104a**. Protrusions **6132a**, **6132b**, may extend approximately parallel to a central longitudinal axis of actuator **6102** and may be arranged around an entire circumference of first flange portion **6104a**. Protrusions **6132a**, **6132b** may extend between a distalmost edge of first flange portion **6104a** and a proximal, rounded edge of first flange portion **6104a**. For example, protrusions **6132a**, **6132b** may terminate distal to the proximal, rounded edge of first flange portion **6104a**. Although second flange portion **6104b** is illustrated as having a smooth surface, second flange portion **6104b** may alternatively include sets of thin, linear protrusions **6132a** and/or protrusions **6132b**.

[0111] In some examples, protrusions **6132b** may align with features of main bodies **2100**, **3100** to indicate to an operator how much main bodies **2100**, **3100** have been rotated relative to first port **2009**, **3009**, respectively. In an example in which actuator **6102** is coupled to main body **2100**, protrusions **6132b** may align with detents (not shown) at an intersection (e.g., a connection point) of main body **2100** and first port **2009**. The intersection of main body **2100** and first port **2009** may include two detents spaced 180 degrees apart and configured to engage with respective slots to lock a position of main body **2100** relative to first port **2009**. Aligning protrusions **6132b** with the detents may assist an operator keep track of how much main body **2100** has been rotated relative to first port **2009**, e.g., 90 degrees, 180 degrees, etc. It will be appreciated that actuators **2102**, **3102**, **4102**, **5102** may include protrusions **6132b** and/or other rotational markers.

[0112] It will be apparent to those skilled in the art at various modifications and variations may be

made in the disclosed devices and methods without departing from the scope of the disclosure. Other aspects of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the features disclosed herein. It is intended that the specification and embodiments be considered as exemplary only.

Claims

1. A medical device comprising: a shaft extending from a proximal end to a distal end, wherein a control wire extends through a lumen of the shaft and to an end effector at the distal end of the shaft; and an operation portion coupled to the shaft, the operation portion including: a housing; one or more attachment devices for coupling the medical device to another medical device; and an actuation mechanism for actuating the end effector, wherein the actuation mechanism includes a lever extending through an opening of a pivotable body within the housing, and wherein the control wire is coupled to the pivotable body.
2. The medical device of claim 1, wherein the medical device further includes a port assembly configured to couple the shaft to the operation portion, and wherein the port assembly is rotatable relative to the operation portion.
3. The medical device of claim 2, wherein the port assembly includes two channels in fluid communication with the lumen of the shaft, and wherein one channel of the two channels is configured to receive the control wire.
4. The medical device of claim 3, wherein another channel of the two channels is configured to couple to a fluid source.
5. The medical device of claim 3, wherein the pivotable body includes a protrusion configured to align with the one channel of the two channels.
6. The medical device of claim 1, wherein the housing includes a first slot on an outer surface of the housing and a second slot inside the housing, and wherein the lever extends through the first slot and the second slot.
7. The medical device of claim 6, wherein the lever is movable along the first slot and the second slot, and relative to the housing.
8. The medical device of claim 6, wherein the first slot and the second slot extend parallel to one another.
9. The medical device of claim 1, wherein the pivotable body includes a channel configured to receive the control wire.
10. The medical device of claim 9, wherein the channel includes a first portion and a second portion extending approximately parallel to one another, and wherein the channel includes a third portion extending between and approximately perpendicular to the first portion and the second portion.
11. The medical device of claim 10, wherein the pivotable body includes a protrusion, and wherein the second portion of the channel extends along the protrusion.
12. The medical device of claim 9, wherein the control wire is fixed to the channel.
13. The medical device of claim 9, wherein the channel includes a first end and a second end, and wherein a second end of the channel is configured to align with the lumen of the shaft.
14. The medical device of claim 1, wherein the actuation mechanism is a first actuation mechanism configured to move the control wire in a proximal direction or a distal direction, and wherein the medical device include a second actuation mechanism configured to rotate the control wire.
15. The medical device of claim 14, wherein the second actuation mechanism includes a first wheel rotatably coupled to a second wheel, wherein the control wire extends through the first wheel, and wherein the second wheel is coupled to a knob.
16. A medical device comprising: a shaft, wherein a control wire extends through a lumen of the shaft and to an end effector at a distal end of the shaft; and an operation portion coupled to the

shaft, the operation portion including: a housing configured to be coupled to a handle of a scope device; and an actuation mechanism coupled to the control wire for actuating the end effector, wherein the actuation mechanism includes an actuator, a first arm, and a second arm movably coupled to one another.

17. The medical device of claim 16, wherein a first end of the first arm is coupled to the actuator and a second end of the first arm is coupled to a first end of the second arm, and wherein a second end of the second arm is coupled to the control wire.

18. The medical device of claim 16, wherein the first arm is configured to move in an axial direction and the second arm is configured to rotate.

19. A medical device comprising: a shaft including an end effector at a distal end of the shaft, wherein a control wire extends through the shaft and is coupled to the end effector; an operation portion including: a housing configured to couple to a handle of another medical device; and an actuation mechanism for actuating the end effector, wherein the actuation mechanism includes a knob, a rod, and a pair of wheels, wherein the rod extends from the knob to a first wheel of the pair of wheels, and wherein the control wire extends through and is rotatably fixed to a second wheel of the pair of wheels.

20. The medical device of claim 19, wherein a rotational axis of the first wheel is transverse to a rotational axis of the second wheel.
